CSCI 171 - Introduction to Programming

Fall 2026

August 24 - December 11

Monday, Wednesday, Friday, 2:20-4:10 p.m.

Instructor

Instructor: Anthony Ackah Nyanzu

Email: Anthony.AckahNyanzu@Principia.edu

Office hours: Monday and Friday, 1:00-2:00 p.m., and by appointment

Canvas: official grades, due dates, quizzes, and feedback

GitHub: course code, setup instructions, starter files, and released solutions

Course Text

Official textbook: *How to Think Like a Computer Scientist: Learning with Python 3*

Authors: Peter Wentworth, Jeffrey Elkner, Allen B. Downey, and Chris Meyers

Cost: free online

Link: https://openbookproject.net/thinkcs/python/english3e/

Optional reference: OpenStax, *Introduction to Python Programming*.

Please bring a laptop to every class. Email the instructor before class if you do not have one. A notebook is also useful because programming is partly thinking on paper.

Course Description

This course introduces programming using Python, with attention to practical problem solving, creative exploration, debugging, and clear program design. No prior programming experience is required.

Students will learn to read, modify, debug, and write programs using variables, expressions, conditionals, loops, functions, strings, lists, tuples, modules, files, events, algorithms, and introductory objects.

Most programming work happens during class. Group activities give you practice and support; individual projects show what you can do on your own.

Learning Outcomes

By the end of the course, students should be able to:

- Explain how Python programs run and how errors can be used as debugging clues.
- Use variables, expressions, input, output, and basic types to build small programs.
- Use conditionals and loops to control program behavior.
- Write functions that organize code, reduce repetition, and return useful results.
- Work with strings, lists, tuples, files, and modules.
- Build small interactive programs using events and turtle graphics.
- Read unfamiliar code and predict what it will do.

- Test, debug, revise, and explain programs clearly.
- Use GitHub and Canvas to manage course work professionally.
- Be ready for the next programming course, including data structures.

How This Class Works

Programming can feel confusing at first. That does not mean you are bad at it. It means your brain is learning a new kind of precision.

Our weekly pattern is meant to make the course predictable:

- Monday: new topic, condensed slides, code reading, live coding, and a short game, puzzle, or unplugged activity.
- Wednesday: ad hoc group programming build. Odd weeks use instructor-assigned groups; even weeks allow student-chosen groups.
- Friday: textbook connection, short quiz or concept check, and an individual in-class programming project.
- Take-home extension: optional extra credit connected to the Friday individual project.

Group work is practice, not a substitute for individual understanding. You may talk, design, debug, and compare ideas with classmates during group builds. Individual projects must represent your own understanding.

Main Tools

- Python 3
- Thonny during the first part of the semester
- Visual Studio Code beginning around Week 5
- GitHub for course repositories and code submission
- Canvas for official assignment submission, quizzes, rubrics, due dates, and grades

We are not using GitHub Classroom. Course repositories will be organized in the `principia-business-cs` GitHub organization.

Major Course Topics

Week	Dates	Main Topic	Textbook
1	Aug 24-28	Programs are instructions	Ch. 1
2	Aug 31-Sep 4	Variables, expressions, and statements	Ch. 2
3	Sep 7-11	Decisions and conditionals	Ch. 5 intro
4	Sep 14-18	`while` loops and sentinel values	Ch. 7 intro
5	Sep 21-25	`for` loops and turtle patterns	Ch. 7 + Ch. 3 selectively
6	Sep 28-Oct 2	Functions and decomposition	Ch. 4
7	Oct 5-9	Fruitful functions and tests	Ch. 6
8	Oct 12-16	Strings and text processing	Ch. 8
9	Oct 19-23	Lists and collections	Ch. 11
10	Oct 26-30	Tuples and grouped values	Ch. 9

Week	Dates	Main Topic	Textbook
11	Nov 2-6	Modules and useful libraries	Ch. 12
12	Nov 9-13	Files and persistence	Ch. 13
13	Nov 16-20	Events and interactive programs	Ch. 10
14	Nov 23-27	Algorithms and project checkpoint	Ch. 14
15	Nov 30-Dec 4	Objects and classes	Ch. 15
16	Dec 7-11	Objects working together and final demos	Ch. 16

Calendar notes:

- Monday, September 7: Labor Day, no class.
- Saturday-Tuesday, September 26-29: Fall Break, no class.
- Wednesday-Sunday, November 25-29: Thanksgiving Break, no class.
- Wednesday-Friday, December 9-11: final examination days.

Assignments and Grading

Category	Weight
Friday individual in-class projects	30%
Programming assignments extending Friday work	20%
Wednesday group builds and reflections	15%
Quizzes and concept checks	10%
Midterm practical exam	10%
Final project or final practical	10%
Professional practice	5%

Professional practice includes preparation, attendance habits, careful submission, meaningful participation, respectful collaboration, and use of course tools.

Letter grades follow this scale:

Percentage	Grade
92.5 and above	A
89.5-92.4	A-
86.5-89.4	B+
82.5-86.4	B
79.5-82.4	B-
76.5-79.4	C+
72.5-76.4	C

Percentage	Grade
69.5-72.4	C-
66.5-69.4	D+
62.5-66.4	D
59.5-62.4	D-
Below 59.5	F

Quizzes

Quizzes are designed to take about 20 minutes, with up to 30 minutes for students who need more reading time. They cover vocabulary, code reading, and small programming ideas from the slides, textbook, and recent class work.

The slides are condensed, but they are not a replacement for practice. If you only read the slides, you should understand the big ideas. If you also read and practice from the textbook, you will be much stronger.

Programming Projects

Individual Friday projects are completed in class and submitted through Canvas. Some projects may also use GitHub folders or starter files.

Take-home extensions are optional extra credit. They are meant to stretch the Friday project, not replace it. Completed solution examples may be released after submission deadlines so you can compare approaches and learn from them.

Group Builds

Group builds happen on Wednesdays. Groups will change throughout the semester so you practice communicating with different classmates.

For group builds, the goal is shared learning. Usually each student should keep their own copy of the group code or notes, even when the group talks through one solution together. Some weeks may ask for a short reflection instead of a full code submission.

Attendance

This course depends on active, in-person work. Absences, including athletic contests and field trips, may be excused if you receive permission at least one week in advance from the instructor when possible. Otherwise, missed quizzes, tests, in-class projects, and classwork may receive a zero.

Every occurrence of unexcused absence or tardiness after the first three may result in a 5% reduction in the final course grade.

If the instructor has not arrived or contacted the class, students may leave 10 minutes after the normal class starting time.

Class Etiquette

Use laptops for course work during class. Phones, unrelated browsing, games, messaging, and other distractions should be put away unless the instructor gives permission for a class activity.

Be generous with classmates. A good programming partner explains, listens, asks questions, and does not take over the keyboard forever.

Late Work

Late work is generally accepted within 24 hours of the original deadline for a 50% penalty, unless an assignment says otherwise. Because many projects are completed in class, missing class can also mean missing the supported work time.

Talk with the instructor early if something serious is getting in the way. Early communication creates more options.

AI Tool Policy

Your submitted work should represent your thinking, skill, and knowledge. Do not represent another person's work, or work created by generative AI, as your own unless an assignment explicitly allows it.

In this course, AI tools may sometimes be used for explanation, review, or debugging practice when the assignment says so. If AI use is allowed, you must follow the assignment directions and be ready to explain every line of submitted code. Code you cannot explain is not finished work.

Academic Integrity

A Principian is expected to pursue a life of integrity. Academic honesty is essential to a Principia College education.

Academic misconduct includes, but is not limited to, plagiarism, collusion, cheating, and assisting another student in violating academic integrity. Minor cases of poor scholarship may be resolved between instructor and student. More serious cases may be reported according to College policy.

Academic Engagement

Academic engagement requires commitment to deep learning. The instructor will provide materials, instruction, opportunities for practice, feedback, and information to help you assess your progress.

If your academic engagement is headed in an ineffective direction, the instructor may speak with you or send an Academic Engagement Alert inviting you to make a plan for getting back on track. Asking for help is a sign of strength.

Accessibility and Support

We believe every student has the potential to succeed. If you have or suspect you may have a disability that affects your learning, contact the Academic Success Coordinator to discuss options for removing barriers to learning. You may also ask the instructor for help connecting with support.

Student support resources include Academic Technology, the Advising and Academic Success Center, counseling and life coaching, Student Health and Safety, Information Technology, International Student Program and Services, the Library, Spiritual Support, and the Writing Center.

Students facing challenges with basic needs, mental health, or other serious concerns are encouraged to speak with the instructor and connect with Student Life or appropriate campus support services.

Inclement Weather

If Principia cancels classes because of weather, the instructor will communicate through Canvas about course expectations and any rescheduled graded activities. Students are expected to keep up with course work during a cancellation unless Canvas instructions say otherwise.

Final Exam

For the Fall 2026 final examination schedule, the course time slot is MW2: Monday/Wednesday, 2:20-4:10 p.m. The scheduled final exam period is Thursday, December 10, 8:00-10:00 a.m.

Students are required to attend scheduled finals for the classes they are taking unless they receive official approval for a change. Requests to reschedule a final exam must follow College policy and deadlines.

Syllabus Changes

The instructor reserves the right to make changes to this syllabus if doing so supports the educational development of the class. Changes will be announced as soon as possible through Canvas and/or class communication.